

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular & Supplementary Winter Examination-2023

Course: B.Tech.

Branch: Mechanical Engineering/Mech(Sandwich)

Subject Name: Heat Transfer (Code: BTMC501)

Semester: V

Max Marks: 60

Date of Exam: 01 Jan 2024

Duration: 3 hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Attempt Any Two of the following.		12
A)	Derive an equation for heat transfer rate through a hollow cylinder by applying Fourier's Law of heat conduction. The hollow cylinder has: • length L, • inner radius r_i, outer radius r_o • inner surface temperature T_i and outer surface temperature T_o, thermal conductivity k. $T_i > T_o$	(L4/CO1)	6
B)	An insulating powder is densely packed in the annular space between two concentric spheres with radii 75 mm and 50 mm. The inner sphere is uniformly heated with electric power input of 30 W. Steady state temperature attained by inner sphere is 120°C and that by outer sphere is 30°C. <i>Neglect thermal resistance offered by the material of the spheres.</i> a) Calculate thermal conductivity of the powder. b) Calculate the thermal resistance offered by insulating powder.	(L3/CO1)	6
C)	Explain the mechanism of heat conduction. Explain “why is a good conductor of electricity also a good conductor of heat?”	(L4/CO1)	6
Q.2	Attempt Any Two of the following.		12
A)	A solid steel ball 5 cm in diameter and initially at 450°C is quenched in a controlled environment at 90°C with convective heat transfer coefficient of 115 W/m ² K. Determine the time taken by center of the ball to reach a temperature of 150°C. Use following properties of steel ball $c_p = 420 \text{ J/kg.K}$, $\rho = 8000 \text{ kg/m}^3$, $k = 46 \text{ W/m.K}$. Justify the equation used for temperature distribution.	(L3/CO2)	6
B)	State general heat conduction equation in Cartesian coordinate system and write down the situations/conditions for its use.	(L2/CO2)	6

C)	Calculate fin efficiency and heat loss from a fin (<i>assuming insulated fin-tip</i>) with circular cross-section under following conditions: Air temperature : 27°C Temperature of fin at the base : 227°C Thermal conductivity of a fin : 220 W/m K h for a cylindrical surface of a fin : 15 W/m ² K Fin diameter : 1 cm Fin length : 3 cm	(L4/CO3)	6
Q. 3 Attempt Any Two of the following.			12
A)	A hot rectangular plate 5 cm x 3 cm maintained at 200°C is exposed to still air at 30°C. Calculate convective heat transfer rate by natural convection from both sides of the plate if the smaller side of the plate is held vertical. Use following correlation: $Nu = 0.59 Ra^{0.25}$ Use following Air properties at mean film temperature of 115°C Density = 0.91 kg/m ³ , $c_p = 1.009$ kJ/kg K, $\mu = 22.65 \times 10^{-6}$ N.s/m ² , $k = 0.0331$ W/m K	(L4/CO4)	6
B)	Sketch laminar and turbulent boundary layers (BL) for flow over a flat plate.	(L4/CO4)	6
C)	Natural convection heat transfer is governed by Nusselt number, Grashoff number and Prandtl number. State equations for these dimensionless numbers.	(L3/CO4)	6
Q.4 Attempt Any Two of the following.			12
A)	Derive expression for mean temperature difference (LMTD) for parallel flow heat exchanger.	(L4/CO5)	6
B)	Sketch the temperature variations along the length of the heat exchangers with following flow arrangements: 1. Parallel flow 2. Counter flow Write down equation for Overall Heat Transfer Coefficient for a tube-in-tube heat exchanger considering two convective resistances on two sides of a single-layer inner-tube and considering conductive resistance offered by the tube material.	(L3/CO5)	6
C)	In a double-pipe counter-flow heat exchanger 10,000 kg/h of an oil having a specific heat 2095 J/kg.K is cooled from 80°C to 50°C by 8,000 kg/h of cooling water at 25°C. Assume overall heat transfer coefficient of 300 W/m ² K and specific heat of water as 4180 J/kg.K. a) Determine LMTD b) Determine area of the heat exchanger.	(L4/CO5)	6
Q. 5 Attempt Any Two of the following.			12
A)	State any four shape factor relations (algebra) used in radiation heat transfer.	(L2/CO6)	6

B)	State the any two laws of radiation heat transfer.	(L2/CO6)	6
C)	Two large parallel plates, held at temperature 727°C and 227°C, have emissivity values of 0.9 and 0.6 respectively. • Calculate the heat transfer rate by radiation per square meter of plate area (<i>without radiation shield</i>). • Calculate the heat transfer rate by radiation per square meter of plate area if a radiation shield having emissivity 0.4 is placed between the two plates. • Calculate the % reduction in heat transfer rate by use of a radiation shield.	(L3/CO6)	6
	*** End ***		

BL – Bloom’s Taxonomy Levels (L1- Remembering, L2- Understanding, L3 – Applying, L4 – Analyzing, L5 – Synthesizing, L6 - Evaluating)
CO – Course Outcomes